

Difference-in-Differences with Multiple Time Periods*

Brantly Callaway[†]

Pedro H. C. Sant'Anna[‡]

December 1, 2020

Abstract

In this article, we consider identification, estimation, and inference procedures for treatment effect parameters using Difference-in-Differences (DiD) with (i) multiple time periods, (ii) variation in treatment timing, and (iii) when the “parallel trends assumption” holds potentially only after conditioning on observed covariates. We show that a family of causal effect parameters are identified in staggered DiD setups, even if differences in observed characteristics create non-parallel outcome dynamics between groups. Our identification results allow one to use outcome regression, inverse probability weighting, or doubly-robust estimands. We also propose different aggregation schemes that can be used to highlight treatment effect heterogeneity across different dimensions as well as to summarize the overall effect of participating in the treatment. We establish the asymptotic properties of the proposed estimators and prove the validity of a computationally convenient bootstrap procedure to conduct asymptotically valid simultaneous (instead of pointwise) inference. Finally, we illustrate the relevance of our proposed tools by analyzing the effect of the minimum wage on teen employment from 2001–2007. Open-source software is available for implementing the proposed methods.

JEL: C14, C21, C23, J23, J38.

Keywords: Difference-in-Differences, Dynamic Treatment Effects, Doubly Robust, Event Study, Variation in Treatment Timing, Treatment Effect Heterogeneity, Semi-Parametric.

*First complete version: March 23, 2018. A previous version of this paper has been circulated with the title “Difference-in-Differences with Multiple Time Periods and an Application on the Minimum Wage and Employment”. We thank the Editor, the Associate Editor, two anonymous referees, Stéphane Bonhomme, Carol Caetano, Sebastian Calonico, Xiaohong Chen, Clément de Chaisemartin, Xavier D’Haultfoeuille, Bruno Ferman, John Gardner, Andrew Goodman-Bacon, Federico Gutierrez, Sukjin Han, Hugo Jales, Andrew Johnston, Vishal Kamat, Qi Li, Tong Li, Jason Lindo, Catherine Maclean, Matt Masten, Magne Mogstad, Tom Mroz, Aureo de Paula, Jonathan Roth, Donald Rubin, Bernhard Schmidpeter, Yuya Sasaki, Na’Ama Shenhav, Tymon Słoczyński, Sebastian Tello-Trillo, Alex Torgovitsky, Jeffrey Wooldridge, Haiqing Xu and several seminar and conference audiences for comments and suggestions. Code to implement the methods proposed in the paper is available in the R package `did` which is available on CRAN.

[†]Department of Economics, University of Georgia. Email: brantly.callaway@uga.edu

[‡]Department of Economics, Vanderbilt University. Email: pedro.h.santanna@vanderbilt.edu

average treatment effect estimators. Thus, our proposed confidence bands are arguably more suitable for visualizing the overall estimation uncertainty than more traditional pointwise confidence intervals.

We illustrate the practical relevance of our proposal by analyzing the effect of the minimum wage on teen employment. Here, we follow much empirical work on the effects of the minimum wage and exploit having access to panel data and variation in treatment timing across states (e.g., [Card and Krueger \(1994\)](#); [Neumark and Wascher \(2000, 2008\)](#); [Dube et al. \(2010\)](#), among many others) in order to estimate the effect of the minimum wage on employment. Interestingly, in our setup, using our approach leads to qualitatively different results than results from the TWFE estimator. This suggests that, at least in certain applications, using methods that are robust to treatment effect heterogeneity can lead to meaningful differences relative to more standard TWFE regressions.

Recent Related Literature: This paper is related to the recent and emerging literature on heterogeneous treatment effects in DiD and/or event studies with variation in treatment timing; see, e.g., [de Chaisemartin and D’Haultfoeulle \(2020\)](#), [Goodman-Bacon \(2019\)](#), [Imai et al. \(2018\)](#), [Borusyak and Jaravel \(2017\)](#), [Athey and Imbens \(2018\)](#) and [Sun and Abraham \(2020\)](#). All these papers present, among other things, some negative results about the interpretation of parameters associated with standard TWFE linear regression specifications; see also [Laporte and Windmeijer \(2005\)](#), [Wooldridge \(2005a\)](#), [Chernozhukov et al. \(2013\)](#), and [Gibbons et al. \(2018\)](#) for earlier related results based on (one-way) fixed-effect estimators. Our proposed procedure completely bypasses the pitfalls highlighted in these papers as we clearly separate the identification, aggregation and estimation/inference steps of the analysis.

These aforementioned papers also propose alternative DiD estimators that do not suffer from the pitfalls associated with TWFE. Among these, perhaps the closest to our proposal are those of [de Chaisemartin and D’Haultfoeulle \(2020\)](#), and [Sun and Abraham \(2020\)](#), though several major differences are worth stressing.

[de Chaisemartin and D’Haultfoeulle \(2020\)](#) is focused on recovering an instantaneous treatment effect measure, while we pay particular attention to treatment effect dynamics. In fact, our framework allows one to form *families* of different aggregate parameters in a unified manner. Second, while we pay particular attention to the role played by pre-treatment covariates, [de Chaisemartin and D’Haultfoeulle \(2020\)](#) mainly focus on unconditional DiD designs. On the other hand, the setup in [de Chaisemartin and D’Haultfoeulle \(2020\)](#) is more general than ours as we consider staggered adoption designs and they allow for more general treatment selection. Nonetheless, we note that the unconditional versions of our parallel trends assumptions are weaker than the one in [de Chaisemartin and D’Haultfoeulle \(2020\)](#), even if one were to specialize their setup to staggered adoption designs.

[Sun and Abraham \(2020\)](#) proposes a parameter, cohort-specific average treatment effects, that translates our group-time average treatment effects from calendar time into event time. [Sun and Abraham \(2020\)](#) proposes regression-based estimators of these parameters that have similar properties to our estimators in the specific case of staggered treatment adoption under an unconditional version of the parallel trends assumption. However, our approach is more general in several respects. First, we allow for parallel trends assumptions to hold after conditioning on covariates, and it is not clear how to adapt the regression based estimators in [Sun and Abraham \(2020\)](#) to this case. Second, we consider a wide variety of possible aggregations of group-time average treatment effects where [Sun and Abraham \(2020\)](#) focuses particularly